

CODED — Cloud-Optimized Distributed Earth Data

Benchmarking **IPFS** as a resilience layer for cloud-optimized geospatial data (Zarr · Icechunk · xarray)

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Objective

Decentralized storage networks like **IPFS** have offered content-addressed, location-independent data access for over a decade. Meanwhile the geoscience community has converged on **cloud-optimized formats** — Zarr, and now Icechunk — that make huge environmental datasets readable through efficient chunk-level range requests.

CODED asks one practical question: can IPFS serve as a **resiliency layer for cloud-optimized geospatial data — and what are the real **performance trade-offs**?**

Approach

- Published a **500-hour ERA5 2-m air-temperature** slice (2.08 GB logical, 500 Zarr chunks) to IPFS via Kubo.
- `xarray.open_zarr()` reads directly over an IPFS gateway — **no code changes**, just point at the CID.
- Benchmarked the **identical dataset** three ways: Icechunk on S3, IPFS HTTP gateway, and a local IPFS daemon over Bitswap.
- All three return a **bit-identical** result (area-weighted mean = **286.5062 K**) — verifying correctness alongside speed.

Bottom Line

IPFS works as a resilience layer — and the access path is everything. An HTTP gateway co-located with the data delivers S3-class throughput (even beating native S3 here).

But cold peer-to-peer **Bitswap** replication across a high-latency link is glacial. Front your IPFS node with a regional gateway, and content-addressing buys you verifiable, takedown-resistant data either way.

The Singapore Test

To stress the network path, a fresh client in **Singapore** (`ap-southeast-1`, ~221 ms RTT) read the full 2.08 GB field from three back-ends served in **us-east-1**.

Backend (cold read)	Time	Throughput
IPFS HTTP gateway	12.5 s	166.6 MB/s
Icechunk on S3	18.4 s	112.8 MB/s
IPFS local daemon (Bitswap)	655 s	3.2 MB/s

Full 2.08 GB cold read, Singapore → us-east-1. The co-located IPFS gateway beat native S3 by ~50%; cold Bitswap was ~52× slower than the gateway.

1.5×
IPFS gateway faster than S3

52×
cold Bitswap slower than the gateway

Tuning Bitswap

Raising the Kubo chunker from 1 MiB to the 2 MiB ceiling cut blocks-per-chunk from 4 → 3, reducing Bitswap round-trips over the Pacific link.

655→312s
cold Bitswap read after chunker tuning

2.1×
speedup on the P2P path

Even so, Bitswap stayed ~23× behind the HTTP gateway — the gap is session-management overhead, not block size.

What This Means

- Gateways win for consumers.** Reading through any HTTP gateway in the data's region gives S3-class (or better) throughput with zero code changes.
- Cold P2P replication is slow** over high-RTT links with a single peer — budget ~1 MB/s; more peers and tuning help.
- Resilience holds regardless.** Content-addressing + pinning keeps datasets verifiable and alive as nodes come and go.